

Lesson 30 — Story Studio (Design Challenge)

Matches LEGO Teacher Portal lesson: "Story Studio".

Objective (I can...): I can design an interactive story that uses AI to respond to the audience. **CS concept:** Applying trained classifiers with events, conditionals, loops, and variables · **Builds on:** Lessons 26-29 and the whole course. *Privacy: on-device camera, no logins, images never saved. Also serves as flex/catch-up — use today to finish and polish anything outstanding from the AI unit.*

AI Connection (2 min): Today your build senses, decides, counts, and responds to an AI YOU trained — every unit of the year working together. *First-principles question:* "Which idea from this year did your story rely on most?"

Materials: LEGO CS & AI kit + device per pair with camera, plan sheets, story props/cards. **Model already built** — reuse the pose-reactor (L28) or Monster model (L26), or a small reconfig; the 30 min is plan + code + test, not build-from-scratch.

60-minute flow - Warm-up (8): Story spark — pairs pitch a one-sentence story where the audience's pose or object changes what happens next. - **Concept + demo (10):** Challenge: build an interactive story where an AI recognition (pose or object) drives the model and the story. Coach models a short plan connecting a recognition to a story beat. Restate privacy points. - **Build/Code (30):** Pairs plan and code an interactive story using a classifier (pre-trained or their own) plus at least one conditional and, ideally, a variable or loop, on the already-built model. *Swap at 15 min.* - **Challenge (7):** Add a branching path so a different recognition leads to a different ending. - **Share + pack-up (5): Story film festival** — each pair performs their interactive story while the coach records a short video clip for the parent note. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator directs the story beats and cues recognitions; swap at 15 min. (*Trio rotates plan/code/perform.*)

Differentiation: *Support* — one recognition, one story beat, partial plan. *Extension / fast-finisher* — multiple recognitions with branching endings and a score. *Reteach* — reuse their Lesson 28-29 trained classifier and add just one story reaction.

Exit check: Pair delivers a working interactive story that uses AI recognition to change what happens, with a matching plan. **Progress layer:** Send home the **Unit 6 (course finale) parent note** ("your child can train an AI and build a machine that senses, decides, counts, and responds") with the **build photo** — the year's capstone. **At-home (optional):** Storyboard a 3-scene interactive story; note where the audience chooses the path. ~10 min.

Standards Crosswalk

Aligned to the **CSTA K-12 CS Standards** — primarily **Algorithms & Programming (1B-AP)**, with connections to **Data & Analysis (1B-DA)** and **Impacts of Computing (1B-IC)**. This is a targeted map, not a claim of full coverage. *Maps to the Virginia CS Standards of Learning, grades 3-5 (confirm exact SOL numbers).*

Unit (Lessons)	Skills taught	CSTA 1B codes targeted
1 · Basics (1-5)		

Unit (Lessons)	Skills taught	CSTA 1B codes targeted
	Sequencing, decomposition, pseudocode/algorithm planning, abstraction (plain-words plan)	1B-AP-11 (decompose problems into smaller parts); 1B-AP-10 (create programs with sequences); 1B-AP-08 (compare/refine multiple solutions — surfaced at design challenges)
2 • Events (6–10)	Event triggers; buttons, taps, and sensor inputs; multiple event handlers	1B-AP-10 (events/control structures); 1B-AP-11 (decomposition)
3 • Loops (11–15)	Repeat loops, forever/stop, nested loops (<i>stretch</i>), abstraction (a loop hides copied blocks), debugging	1B-AP-13 (use loops/iteration); 1B-AP-15 (test and debug); 1B-AP-10
4 • Conditionals (16–20)	If/then, true/false, if/then/else, sensor-threshold decisions; comparing two working solutions	1B-AP-10 (conditionals/control); 1B-AP-15 (debug); 1B-AP-08 (compare/refine solutions)
5 • Variables (21–25)	Create/store a variable, increment/counting, variables in decisions, multiple variables	1B-AP-09 (create programs that use variables); 1B-AP-10
6 • AI & Data (26–30)	Use a pre-trained classifier; train a custom classifier; examples as balanced data; accuracy, error, and fairness	1B-DA-06 (data used to make predictions / examples as training data); 1B-IC-18 (impacts, fairness, and who is helped or harmed); 1B-AP-10
Pair programming (every lesson)	Driver/Navigator roles, swapping, collaboration	1B-AP-16 (take on varying roles when collaborating)
Cross-cutting: debugging mini-tasks (L12, L14, L17, L19) reinforce 1B-AP-15 ; the "two ways to solve it" and design-challenge share-outs reinforce 1B-AP-08 ; the AI-impacts prompts (L8, L13, L18, L25) and L29 fairness discussion reinforce 1B-IC-18 .		